Definition (5.29)

Let f be continuous on $[a, b]$. The **average value** f_{av} of f on $[a, b]$ is

$$f_{av} = \frac{1}{b-a}\int_a^b f(x)\,dx.$$

Note that, by the mean value theorem for definite integrals, if f is continuous on $[a, b]$, then

$$f_{av} = f(z) \quad \text{for some } z \text{ in } [a, b].$$

EXAMPLE 6 Given $\int_0^3 x^2\,dx = 9$, find the average value of f on $[0, 3]$.

SOLUTION By Definition (5.29), with $a = 0$, $b = 3$, and $f(x) = x^2$,

$$f_{av} = \frac{1}{3-0}\int_0^3 x^2\,dx = \frac{1}{3}\cdot 9 = 3.$$

In the interval $[0, 3]$, the function values $f(x) = x^2$ range from $f(0) = 0$ to $f(3) = 9$. Note that the function f *takes on* its average value 3 at the number $z = \sqrt{3}$.

EXERCISES 5.5

Exer. 1–6: Evaluate the integral.

1 $\int_{-2}^{4} 5\,dx$ 2 $\int_1^{10} \sqrt{2}\,dx$ 3 $\int_6^2 3\,dx$

4 $\int_4^{-3} dx$ 5 $\int_{-1}^{1} dx$ 6 $\int_2^2 100\,dx$

Exer. 7–10: It will follow from the results in Section 5.6 that

$$\int_1^4 x^2\,dx = 21 \quad \text{and} \quad \int_1^4 x\,dx = \tfrac{15}{2}.$$

Use these facts to evaluate the integral.

7 $\int_1^4 (3x^2 + 5)\,dx$ 8 $\int_1^4 (6x - 1)\,dx$

9 $\int_1^4 (2 - 9x - 4x^2)\,dx$ 10 $\int_1^4 (3x + 2)^2\,dx$

Exer. 11–16: Verify the inequality without evaluating the integrals.

11 $\int_1^2 (3x^2 + 4)\,dx \geq \int_1^2 (2x^2 + 5)\,dx$

12 $\int_1^4 (2x + 2)\,dx \leq \int_1^4 (3x + 1)\,dx$

13 $\int_2^4 (x^2 - 6x + 8)\,dx \leq 0$

14 $\int_2^4 (5x^2 - x + 1)\,dx \geq 0$

15 $\int_0^{2\pi} (1 + \sin x)\,dx \geq 0$

16 $\int_{-\pi/3}^{\pi/3} (\sec x - 2)\,dx \leq 0$

Exer. 17–22: Express as one integral.

17 $\int_5^1 f(x)\,dx + \int_{-3}^5 f(x)\,dx$

18 $\int_4^1 f(x)\,dx + \int_6^4 f(x)\,dx$

19 $\int_c^d f(x)\,dx + \int_e^c f(x)\,dx$

20 $\int_{-2}^6 f(x)\,dx - \int_{-2}^2 f(x)\,dx$

21 $\int_c^{c+h} f(x)\,dx - \int_c^h f(x)\,dx$

22 $\int_c^m f(x)\,dx - \int_d^m f(x)\,dx$

Exer. 23–30: The given integral $\int_a^b f(x)\,dx$ may be verified using the results in Section 5.6. (a) Find a number z that satisfies the conclusion of the mean value theorem (5.28). (b) Find the average value of f on $[a, b]$.

23 $\int_0^3 3x^2\,dx = 27$ 24 $\int_{-4}^{-1} \frac{3}{x^2}\,dx = \frac{9}{4}$

25 $\int_{-2}^{1} (x^2 + 1)\,dx = 6$

26 $\int_{-1}^{3} (3x^2 - 2x + 3)\,dx = 32$

27 $\int_{-1}^{8} 3\sqrt{x+1}\,dx = 54$ 28 $\int_{-2}^{-1} \frac{8}{x^3}\,dx = -3$

29 $\int_1^2 (4x^3 - 1)\,dx = 14$ 30 $\int_1^4 (2 + 3\sqrt{x})\,dx = 20$

C **Exer. 31–32: The given integral may be verified using results in Section 5.6. Use Newton's method to approximate, to three decimal places, a number z that satisfies the conclusion of the mean value theorem (5.28).**

31 $\int_{-2}^{3} (8x^3 + 3x - 1)\,dx = 132.5$

32 $\int_{\pi/6}^{\pi/4} (1 - \cos 4x)\,dx = \dfrac{\pi}{12} + \dfrac{\sqrt{3}}{8}$

33 Let f and g be integrable on $[a, b]$. If c and d are r real numbers, prove that

$$\int_a^b [cf(x) + dg(x)]\,dx = c\int_a^b f(x)\,dx + d\int_a^b g(x)\,dx$$

34 If f is continuous on $[a, b]$, prove that

$$\left|\int_a^b f(x)\,dx\right| \le \int_a^b |f(x)|\,dx.$$

(*Hint:* $-|f(x)| \le f(x) \le |f(x)|$.)

5.6 THE FUNDAMENTAL THEOREM OF CALCULUS

This section contains one of the most important theorems in calculus. In addition to being useful in evaluating definite integrals, the theorem also exhibits the relationship between derivatives and definite integrals. The theorem, aptly called *the fundamental theorem of calculus*, was discovered independently by Sir Isaac Newton (1642–1727) in England and by Gottfried Wilhelm Leibniz (1646–1716) in Germany. It is primarily because of this discovery that both men are credited with the invention of calculus.

To avoid confusion in the following discussion, we shall use t as the independent variable and denote the definite integral of f from a to b by $\int_a^b f(t)\,dt$. If f is continuous on $[a, b]$ and $a \le x \le b$, then f is continuous on $[a, x]$; therefore, by Theorem (5.20), f is integrable on $[a, x]$. Consequently, the formula

$$G(x) = \int_a^x f(t)\,dt$$

determines a function G with domain $[a, b]$, since for each x in $[a, b]$ there corresponds a unique number $G(x)$.

To obtain a geometric interpretation of $G(x)$, suppose that $f(t) \ge 0$ for every t in $[a, b]$. In this case we see from Theorem (5.19) that $G(x)$ is the area of the region under the graph of f from a to x (see Figure 5.28).

As a specific illustration, consider $f(t) = t^3$ with $a = 0$ and $b > 0$ (see Figure 5.29). In Example 7 of Section 5.3 we proved that the area under the graph of f from 0 to b is $\frac{1}{4}b^4$. Hence the area from 0 to x is

$$G(x) = \int_0^x t^3\,dt = \tfrac{1}{4}x^4.$$

This gives us an explicit form for the function G if $f(t) = t^3$. Note that in this illustration,

$$G'(x) = D_x(\tfrac{1}{4}x^4) = x^3 = f(x).$$

Thus, by Definition (5.1), G is an antiderivative of f. This result is not an accident. Part I of the next theorem brings out the remarkable fact that if f is *any* continuous function and $G(x) = \int_a^x f(t)\,dt$, then G is an antiderivative of f. Part II of the theorem shows how *any* antiderivative may be used to find the value of $\int_a^b f(x)\,dx$.

FIGURE 5.28

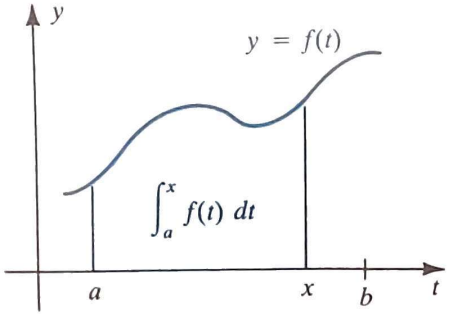


FIGURE 5.29

